ROS1 Application

System Configuration: ubuntu20.04

ROS1 Version: noetic

1. ROS1 Environment Configuration

a. Set upROS1Installation Source

ROS1清华源下载命令

```
sudo sh -c '. /etc/lsb-release && echo "deb http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu/ `lsb_release -cs` main" > /etc/apt/sources.list.d/ros-latest.list'
```

b. Set Key

设置信任 ROS 的 GPG Key，并更新索引

```
sudo apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
```

代码块

```
sudo apt update
```

c. InstallROS1 (Official Download)

ROS1安装命令

```
sudo apt install ros-noetic-desktop-full
```

> 💡 Proxy Accelerated Download and Installation of ROS1
>
> wget http://fishros.com/install -O fishros && . fishros

d. Configure environment variables

```
echo "source /opt/ros/noetic/setup.bash">> ~/.bashrc
```

代码块

```
source ~/.bashrc
```

2. Device Connected to Virtual Machine

a. View Device

代码块

```
ll /dev/ttyUSB*
```

b. Establish Port Mapping

代码块

```
sudo gedit /etc/udev/rules.d/99-serial-imu.rules
```

c. Fill in the content of the mapping file

代码块

```
KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523", MODE:="0777", SYMLINK+="imu-serial"
```

d. Save and exit, then run the command to make the rules take effect

代码块

```
sudo udevadm trigger
```

代码块

```
sudo service udev reload
```

代码块

```
sudo service udev restart
```

e. **Verify**

代码块

```
ll /dev/imu-serial
```

代码块

```
sudo usermod -aG dialout ash
```

3. Import the prepared Compressed Packet

a. **Feishu has it in the same level directory IMU_ROS1.zip**

b. **Transferred into the virtual machine via file transfer software**

c. **Install the IMU_Library**

代码块

```
cd IMU_ROS1
# 下载解压IMU_ROS1压缩文件后，进入到IMU_Library目录下，运行以下指令
cd IMU_Library

# 安装库及其依赖
pip install -e .

# 或使用setup.py安装
python setup.py install
```

4. Installation of Python-related Libraries

代码块

```
sudo pip3 install pyserial
sudo pip3 install smbus2
```

When rendering issues occur, execute the following command

代码块

```
sudo apt-get install ros-noetic-imu-filter-madgwick
sudo apt-get install ros-noetic-rviz-imu-plugin
```

5. Build ROS1 Project

a. Open a new terminal in the /home directory and create a new ROS1 workspace

代码块

```
mkdir imu_ros1
cd imu_ros1
mkdir src
cd src/
catkin_init_workspace
```

b. Copy the transferred file IMU_ROS1 folder to the ~/imu_ros1/src/ directory

代码块

```
# 复制 IMU_ROS1 文件夹到新建的 src 目录下
cp -r ~/IMU_ROS1 ~/imu_ros1/src
cd ~/imu_ros1
catkin_make
```

c. Write the working directory ~/imu_ros1 into the environment variable

代码块

```
# 编辑 ~/.bashrc
sudo gedit ~/.bashrc

# 把下面命令写到末尾
source ~/imu_ros1/devel/setup.bash

source ~/.bashrc
```

6. Start ROS1 Node

a. Open the terminal, enter roscore to start the node

代码块

```
# 启动roscore
roscore

```

```
# 新开终端，设置环境，启动节点
source ~/imu_ros1/devel/setup.bash
```

b. Add executable permission to the Python script (crucial)

Enter the `scripts` directory where the script is located, and execute the `chmod +x` command to grant executable permissions (`+x` means add execute, i.e., add execution permissions):

代码块

```
# 进入imu_driver.py所在目录（按你的实际路径）
cd ~/imu_ros1/src/IMU_ROS1/scripts/

# 赋予可执行权限（仅需执行1次，永久生效）
chmod +x imu_driver.py
chmod +x mag_visualizer.py
```

Return to the imu_ros1 folder and run imu_driver.py

代码块

```
cd ~/imu_ros1
```

代码块

```
rosrun IMU_ROS1 imu_driver.py
```

7. IMU Data Printing

a. Open a new terminal and view the imu topic

代码块

```
# 查看当前发布的所有话题
rostopic list
```

b. Print topic data

代码块

```
# 打印IMU原始数据
rostopic echo /imu/data_raw
```

```

# 打印磁力计数据
rostopic echo /imu/mag
```

8. RViz Visualization

a. **Run the command to start rviz()**

代码块

```
roslaunch IMU_ROS1 imu_display.launch
```

9. Frequently Asked Questions

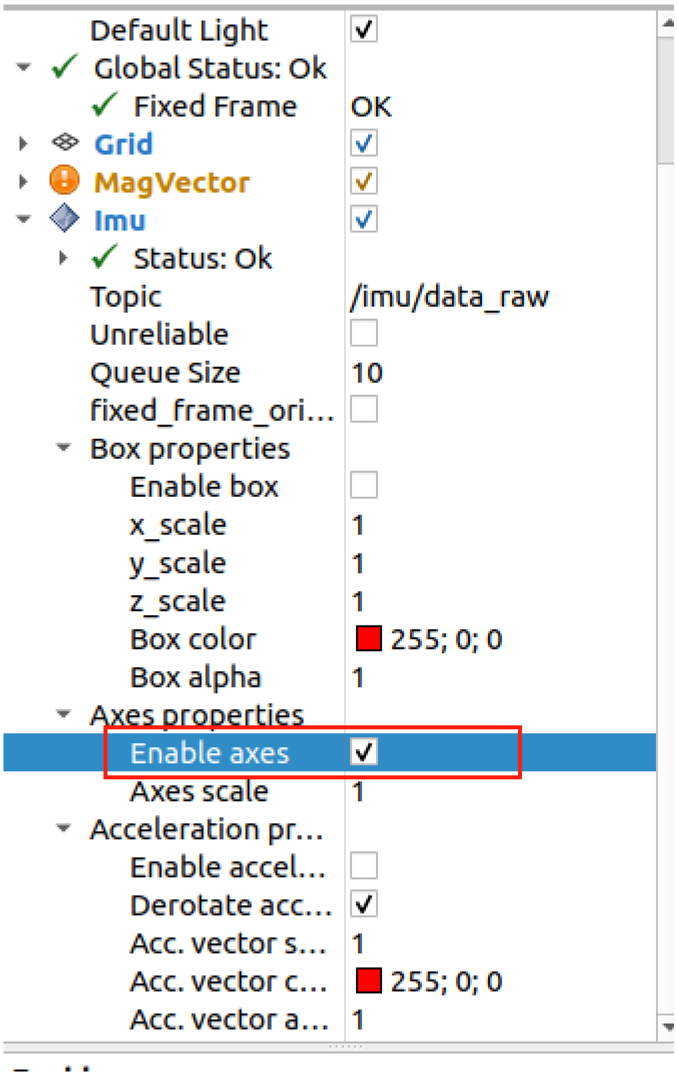
a. If the startup node fails to open, please try the following commands

代码块

```
# 在~/imu_ros1目录下运行
source devel/setup.bash

# 端口号问题
sudo chmod 666 /dev/imu-serial
```

b. The three axes display is very small in RVIZ visualization, recheck Enable axes



Default Light
Global Status: Ok
Fixed Frame
OK
Grid
MagVector
Imu
Status: Ok
Topic
/imu/data_raw
Unreliable
Queue Size
10
fixed_frame_ori...
Box properties
Enable box
x_scale
1
y_scale
1
z_scale
1
Box color
255; 0; 0
Box alpha
1
Axes properties
Enable axes
Axes scale
1
Acceleration pr...
Enable accel...
Derotate acc...
Acc. vector s...
1
Acc. vector c...
255; 0; 0
Acc. vector a...
1